

The Turbulence Identity Problem

A Falsifiable Experimental Criterion

1. The identity problem

The question addressed here is narrower than the general problem of turbulence and stronger than ordinary recognition. It is not how to compute every turbulent statistic, construct a universally satisfactory closure model, predict transition in every geometry, or explain every manifestation of intermittency. It is this: **can one operational criterion identify a broad canonical class of turbulent regimes while excluding neighboring flows that reproduce some of the same familiar signatures?**

That question requires a different standard from descriptive resemblance. A useful identity criterion must survive variation among accepted turbulent flows without being defined by any one of them. It must also be allowed to fail. An accepted turbulent regime that violates a necessary part of the criterion under a valid, adequately powered execution is evidence against the criterion, not an invitation to redefine the regime after the result. Conversely, a nonturbulent comparator that satisfies the complete criterion under the same standard is a sufficiency failure. The positive and negative roles therefore have to be warranted independently of the proposed test and frozen before its outcome is known.

The relevant object is a **regime**, not an instantaneous velocity field and not an isolated invariant solution. This distinction matters because dynamically important recurrent solutions can sit inside turbulent dynamics, reproduce significant turbulent statistics, and be visited or shadowed by turbulent trajectories without thereby exhausting the containing turbulent regime. In plane Couette flow, unstable periodic motions have been found whose temporal statistics closely reproduce features of the turbulent state and around which turbulent trajectories wander before escaping and returning.

Periodic solutions have likewise been extracted from forced isotropic turbulence and compared directly with its statistics. Laboratory and numerical work has also observed

turbulent trajectories transiently shadowing unstable periodic orbits and spending substantial time near them.¹

None of this makes recurrent solutions incidental. It does the opposite: it makes the level distinction unavoidable. If an individual periodic orbit can be dynamically and statistically important to turbulence, then “periodic” cannot by itself mean “not turbulence,” and “turbulence-like statistics” cannot by themselves establish that one orbit is the whole turbulent regime. An identity test therefore has to distinguish the standing of an isolated recurrent realization from the standing of the larger regime that may approach, shadow, leave, and revisit such realizations.

This paper treats identity as an empirical discrimination problem. The proposed criterion is required to answer the same question across maintained and decaying turbulence, homogeneous and inhomogeneous turbulence, wall-free and wall-bounded turbulence, and temporally or spatially developing regimes within a declared physical scope. It must then face comparators chosen precisely because they preserve substantial pieces of what is ordinarily associated with turbulence. A criterion that succeeds only after the difficult comparators are removed is not an identity criterion for the claimed class.

2. Why familiar signatures are not enough by themselves

The difficulty is not that familiar turbulence diagnostics are uninformative. The difficulty is that several of them can be preserved outside the distinction the present experiment is trying to make.

Chaos provides a sharp example. Counter-rotating Taylor–Couette flow has been shown to possess a low-dimensional chaotic set emerging through a Feigenbaum cascade, with unstable periodic orbits of the Navier–Stokes system corresponding to periodic points of a low-dimensional map. In that system, recurrent patterns can

¹ G. Kawahara and S. Kida, “Periodic motion embedded in plane Couette turbulence: regeneration cycle and burst,” *Journal of Fluid Mechanics* 449 (2001): 291–300, doi:10.1017/S0022112001006243; L. van Veen, S. Kida and G. Kawahara, “Periodic motion representing isotropic turbulence,” *Fluid Dynamics Research* 38 (2006): 19–46, doi:10.1016/j.fluiddyn.2005.09.001; B. Suri, L. Kageorge, R. O. Grigoriev and M. F. Schatz, “Capturing Turbulent Dynamics and Statistics in Experiments with Unstable Periodic Orbits,” *Physical Review Letters* 125 (2020): 064501, doi:10.1103/PhysRevLett.125.064501.

reconstruct the statistics of the chaotic fluid motion.² Chaos, recurrence and successful statistical reconstruction can therefore coexist in a fluid system whose dynamical organization is far more constrained than the broad canonical turbulence class considered here. A criterion that reduces turbulence to chaos alone has no remaining resource for making that distinction.

Statistics provide a different pressure. Jiménez constructed reduced stochastic Markov models of turbulent channel-flow projections that can mimic many statistics of the original flow even with relatively crude operator approximations.³ The result is not that turbulence statistics are dispensable. It is that statistical resemblance does not by itself certify the same realized dynamical provenance. A model can reproduce distributions of observables while changing what physically carries the evolution that generated them. For an identity claim, that distinction matters.

The converse pressure appears in reduced interaction models. Generalised quasilinear channel-flow models can recover substantial turbulent behavior while sharply restricting which Fourier groups retain full nonlinear interaction.⁴ More recently, mixed-dynamics simulations of homogeneous isotropic turbulence have reproduced important turbulent properties while allowing only a fraction of the resolved modes to obey the full Navier–Stokes dynamics, with the remainder assigned much simpler evolution.⁵ These results make raw state size, raw mode count and literal retention of every nonlinear interaction poor candidates for the identity quantity. A turbulent field can require many degrees of freedom to represent while the causally load-bearing nonlinear-transfer architecture is a different question.

Taken together, these examples block several tempting shortcuts. Recurrence and chaos do not by themselves identify the regime; statistical resemblance does not guarantee the same dynamical provenance; and large state dimension or literal retention of every nonlinear interaction is not the same thing as causally load-bearing transfer support. The point is not to discard spectra, structures, recurrence, chaos,

² B. Wang, R. Ayats, K. Deguchi, Á. Meseguer and F. Mellibovsky, “Mathematically established chaos and forecast of statistics with recurrent patterns in Taylor–Couette flow,” *Journal of Fluid Mechanics* 1011 (2025): R2, doi:10.1017/jfm.2025.151.

³ J. Jiménez, “Fake turbulence,” *Journal of Fluid Mechanics* 990 (2024): R2, doi:10.1017/jfm.2024.516.

⁴ C. G. Hernández, Q. Yang and Y. Hwang, “Generalised quasilinear approximations of turbulent channel flow. Part 1. Streamwise nonlinear energy transfer,” *Journal of Fluid Mechanics* 936 (2022): A33, doi:10.1017/jfm.2022.59.

⁵ D. Donzis and S. Sajeev, “Degrees of freedom and the dynamics of fully developed turbulence,” *Physical Review Fluids* 9 (2024): 044605, doi:10.1103/PhysRevFluids.9.044605.

statistics or reduced models. It is to require any identity-bearing diagnostic to contribute to a criterion that is necessary across independently accepted turbulent regimes and sufficient against independently accepted false substitutes.

3. A compound candidate

The proposed criterion is therefore compound. Within the declared scope Σ_C , the hypothesis to be tested is

$$\text{turbulence} \Leftrightarrow \text{DLBTF} \wedge \text{HFN}.$$

The equivalence sign is a **candidate empirical equivalence**, not a reported result. **Distributed Load-Bearing Transfer Flow** concerns the physical nonlinear-transfer burden. A regime satisfies the DLBTF side only if its continuation depends on causally load-bearing nonlinear-transfer capacity and that sufficient capacity does not close to one uniformly bounded physical transfer interface as the problem is enlarged along a declared admissible physical axis. “Distributed” does not mean that many grid points, Fourier coefficients or latent coordinates exist. “Load-bearing” requires causal consequence: removing the claimed transfer support must impair the frozen reference-regime burden relative to an appropriate sham, and restoration or competent replacement must be tested rather than assumed. “Opening” refers only to supported growth of the required causal transfer capacity over a declared finite physical enlargement ladder. It does not mean that an infinite support has been mathematically proved.

This distinction makes the negative test unusually strong. If one independently warranted turbulent regime admits a competent, uniformly bounded transfer interface that preserves the complete preregistered regime burden across its physical enlargement ladder, DLBTF necessity fails for that flow case. Searching unsuccessfully for such a closure cannot reverse the inference. Failure to find a bounded closure is evidence only that the tested closures failed. It is not evidence that no bounded closure exists.

The positive side therefore requires a different route. The protocol introduces **Causal Transfer Response Rank**. At a frozen reference state, small normalized interventions are made directly to the nonlinear-transfer contribution, the intervention is removed, and a preregistered vector of future regime-relevant responses is

measured. The resulting local intervention-to-response operator asks how many independent directions of altered nonlinear transfer produce distinguishable future consequences above the permitted error floor. Later sections develop the rank bound and its tolerance-aware form. For the present argument, its job is simple: OPENING must be supported by a positive lower bound on required causal transfer capacity, not inferred from the inability of investigators to invent a successful reduced model.

History-Family Nondegeneracy performs a different job. HFN asks whether an independently warranted regime supports a non-degenerate family of realized Navier–Stokes histories or instead collapses, after a preregistered symmetry, phase and time-shift quotient, onto one dynamically thin recurrent itinerary or finite thin recurrent family. HFN is not synonymous with a positive Lyapunov exponent, aperiodicity, stochastic forcing or large state-space dimension. An unstable recurrent orbit can possess expanding directions while remaining one recurrent orbit. Random forcing can generate different trajectories without showing that the internal dynamics supplies the relevant history plurality. HFN instead requires distinct realized histories that remain within the same independently warranted regime, separate through the regime’s own dynamics, and are not exhausted by the recurrent class against which the test was preregistered.

The two conjuncts are retained separately because they answer separable failure questions. A dynamically diverse chaotic flow can pressure the HFN side while remaining, in principle, physically transfer-closable by a bounded architecture.

Conversely, an isolated exact or recurrent Navier–Stokes solution can carry rich nonlinear transfer while remaining dynamically thin at the level of the isolated solution. Combining transfer complexity and history diversity under one label would not remove either empirical burden; it would only make failures harder to locate.

Each side therefore has a genuine three-way result. The transfer assay returns **BOUNDED**, **OPENING**, or **UNRESOLVED**. The history assay returns **NONDEGENERATE**, **RECURRENT-THIN**, or **UNRESOLVED**. Failure to establish OPENING does not establish BOUNDED, and failure to establish NONDEGENERATE does not establish recurrence collapse. Insufficient power, an unresolved representation conflict, an inadequate enlargement ladder, or an

ambiguous recurrence test remains an epistemic limitation rather than being converted into evidence for or against the identity hypothesis.

This structure gives the proposal direct falsifiers. One independently warranted positive regime returning competent BOUNDED transfer support defeats transfer necessity. One independently warranted positive regime shown to be RECURRENT-THIN at the regime-family level defeats history necessity. Conversely, one independently warranted nonturbulent comparator that competently returns both OPENING and NONDEGENERATE defeats sufficiency. No post-result change of scope, role, fidelity criterion or dynamical level can rescue that adjudication.

4. Declared scope and the adjudication surface

A necessary-and-sufficient claim is only as meaningful as the physical class over which it is asked to hold. The first test is therefore deliberately narrower than “all turbulence.” Its declared scope, Σ_C , is established three-dimensional, single-phase, constant-property, non-rotating, non-stratified, non-buoyant, incompressible Newtonian Navier–Stokes turbulence in canonical homogeneous, free-shear and smooth-wall flow families, away from laminar–turbulent transition. The exclusions matter. Compressible, rotating, strongly stratified or buoyant, rough-wall, strongly curved or separated wall flows, multiphase and reactive flows, magnetohydrodynamic and plasma turbulence, quantum or superfluid turbulence, and generic spatiotemporal chaos are not negative cases for the first experiment. They are outside its jurisdiction. A successful result inside Σ_C would not silently extend to them.

The positive test surface is designed to make the criterion survive substantial variation inside that jurisdiction. Forced homogeneous isotropic turbulence provides a maintained, wall-free reference with extensive direct numerical precedent.⁶ Freely decaying homogeneous turbulence removes continued external maintenance while retaining the homogeneous turbulence problem.⁷ Homogeneous shear turbulence introduces sustained mean-shear production without a solid wall.⁸ Turbulent channel

⁶ Y. Li, E. Perlman, M. Wan, Y. Yang, C. Meneveau, R. Burns, S. Chen, A. Szalay and G. Eyink, “A public turbulence database cluster and applications to study Lagrangian evolution of velocity increments in turbulence,” *Journal of Turbulence* 9 (2008): N31, doi:10.1080/14685240802376389.

⁷ P.-Å. Krogstad and P. A. Davidson, “Freely decaying, homogeneous turbulence generated by multi-scale grids,” *Journal of Fluid Mechanics* 680 (2011): 417–434, doi:10.1017/jfm.2011.169.

⁸ A. Sekimoto, S. Dong and J. Jiménez, “Direct numerical simulation of statistically stationary and homogeneous shear turbulence and its relation to other shear flows,” *Physics of Fluids* 28 (2016): 035101, doi:10.1063/1.4942496.

flow supplies an internal smooth-wall architecture with a well-resolved hierarchy of scales.⁹ A turbulent round jet adds a spatially developing free-shear geometry.¹⁰ A zero-pressure-gradient turbulent boundary layer adds a spatially developing external smooth-wall flow.¹¹ The criterion is not defined by these six flows. They are independently recognized test families across which the same candidate is required to survive.

That distinction between a test surface and a definition is essential. If one of these independently warranted turbulent regimes returns a competent failure of either necessary conjunct, the candidate fails necessity within Σ_C . The flow is not subsequently removed from the scope because it proved inconvenient. Likewise, the fidelity standard applied to it cannot be weakened or reinterpreted after its result is known. This preclassification supplies the independent reference classification for the discrimination test; it does not assume that DLBTF or HFN is true of any case.

The negative surface is deliberately heterogeneous. It includes laminar periodic vortex shedding, steady or traveling exact coherent solutions, periodic or quasiperiodic coherent branches, a low-dimensional chaotic Taylor–Couette case, a synthetic or “fake turbulence” control, and a laminar zero-pressure-gradient boundary layer. These are not intended to form a single alternative class. Their common job is adversarial: each preserves something that an overly permissive turbulence criterion might mistake for identity. Some preserve nonlinear fluid dynamics and coherent time dependence. Some preserve recurrence or chaos. Some reproduce statistics. Some preserve the same broad geometry while changing the dynamical regime. A necessary-and-sufficient criterion has to exclude all of them for reasons that survive the same experimental discipline applied to the positives.

One boundary comparison is especially important. An isolated recurrent solution and the turbulent regime that contains, approaches or shadows recurrent solutions are assigned different physical levels before the candidate is inspected. This prevents a recurrent orbit from being called “the turbulence” when its statistics look persuasive

⁹ M. Lee and R. D. Moser, “Direct numerical simulation of turbulent channel flow up to $Re_\tau \approx 5200$,” *Journal of Fluid Mechanics* 774 (2015): 395–415, doi:10.1017/jfm.2015.268.

¹⁰ D.-h. Shin, R. D. Sandberg and E. S. Richardson, “Self-similarity of fluid residence time statistics in a turbulent round jet,” *Journal of Fluid Mechanics* 823 (2017): 1–25, doi:10.1017/jfm.2017.304.

¹¹ A. G. Balasubramanian, L. Guastoni, P. Schlatter and R. Vinuesa, “Direct numerical simulation of a zero-pressure-gradient turbulent boundary layer with passive scalars up to Prandtl number $Pr = 6$,” *Journal of Fluid Mechanics* 974 (2023): A49, doi:10.1017/jfm.2023.803.

and then called “merely an orbit” when its history family proves thin. The experiment asks separately what is true of the isolated recurrent object and what is true of the containing regime. The level cannot change after the result.

Finally, two within-family holdouts test portability rather than introduce new definitions. One is drawn from an internal smooth-wall family other than the primary channel implementation—turbulent pipe flow or plane Couette flow. The other is drawn from a free-shear family other than the primary round jet—turbulent wake or turbulent mixing layer. These cases remain inaccessible to criterion tuning until the method, tolerances and decision rules are frozen. A candidate that works on the flows used to construct its implementation but fails on an independently held member of the same claimed class has not earned class-level closure.

5. One fidelity burden for every decisive flow

The criterion is not tested against a vague requirement that a modified or reduced system “still looks turbulent.” Each decisive comparison is made under a preregistered reference-regime fidelity specification, denoted β . The particular observables can differ with the physics of the flow case, but β always imposes the same four fidelity requirements: preservation of the relevant budgets, preservation of continuation or development of the reference regime, preservation of the causal-response behavior used by the assay, and preservation of the state-distribution features required to establish that the comparison still represents the same independently warranted physical regime.

This is stricter than matching one spectrum, one mean profile or one probability distribution. A candidate reduced interface that reproduces familiar statistics while corrupting the energy budget has not preserved β . Neither has one that matches the budget while destroying the regime’s downstream or temporal continuation, or one that retains both but changes the responses to the interventions on which the causal test depends. The criterion is allowed to ask different flows for different concrete observables because a round jet and homogeneous isotropic turbulence do not possess identical natural diagnostics. It is not allowed to ask them different epistemic questions.

The symmetry matters most on the negative side. A false substitute cannot be excluded because it was subjected to a stricter fidelity requirement than the positive cases, and a positive case cannot be protected by a looser one. The same four fidelity jobs remain live on both sides. A negative flow that satisfies the complete compound criterion under its preregistered β criterion is a real sufficiency problem. A positive flow that fails a necessary conjunct after satisfying its own preregistered β criterion is a real necessity problem.

Competence is itself preregistered rather than inferred from a convenient result. A representation or assay contributes to adjudication only after its declared validity, power and, where applicable, resolution or convergence checks pass while β is preserved. The concrete β tolerances and observable mappings are fixed during calibration; the broader information-order rule governing method development, calibration and decisive holdouts is specified in Section 9.

6. Testing the transfer side

DLBTF begins with an adversarial question rather than a positive claim: **can the nonlinear-transfer capacity required by an independently warranted turbulent regime be replaced by one uniformly bounded physical transfer interface while β remains satisfied across admissible physical enlargement?**

The word physical constrains both the proposed closure and the enlargement. Increasing numerical resolution, adding basis functions, decreasing the time step or increasing floating-point precision does not by itself enlarge the physical problem. An admissible enlargement changes a declared physical control—such as scale separation or another preregistered parameter—while keeping the flow inside the same independently warranted regime family. The enlargement axis and ladder are fixed for each flow case before decisive data are inspected; an axis that crosses laminar–turbulent transition or changes the case into a different physical family does not count as enlargement of that case. The purpose is to ask whether the causally sufficient transfer support required by the regime remains bounded as more physical transfer structure becomes available, not whether a numerical representation becomes larger because the calculation becomes more accurate.

The bounded-closure challenge is existential. Investigators are encouraged to construct the strongest plausible reduced transfer interface they can. Hidden closure variables, memory terms, nonlocal couplings, stochastic internal states, learned latent dynamics and modified evolution operators all count as part of that interface if they carry information across the causal cut. A nominally low-dimensional closure cannot obtain a bounded verdict by relocating the missing transfer capacity into an uncounted memory, auxiliary state or opaque learned subsystem. The relevant quantity is the complete physical interface through which the future β -relevant response is carried.

If a competent bounded interface survives the entire enlargement ladder while preserving β , DLBTF is BOUNDED for that flow case. BOUNDED is not the trivial statement that finitely many measured ranks have a finite maximum. It requires a competent bounded transfer-interface schema that preserves β across the preregistered enlargement ladder under the frozen closure-accounting rule. That is a direct falsifier of DLBTF necessity for an independently warranted positive flow. No exhaustive search over every conceivable model is required to earn this negative result: one successful bounded construction is enough.

The reverse inference is intentionally unavailable. Ten failed closures, or ten thousand, do not establish OPENING. They establish only that those closures failed. The space of possible physical closures is too large for unsuccessful model construction to serve as a proof that no bounded interface exists. DLBTF therefore needs a positive lower-bound measurement that does not depend on the ingenuity or exhaustion of a model search.

Controlled perturbation studies make such a causal posture experimentally plausible. Localized interventions have been used in three-dimensional isotropic turbulence to identify features whose perturbation has measurable downstream dynamical consequences,¹² and corresponding causal-feature experiments have been developed in turbulent channel flow.¹³ The present proposal does not treat those studies as prior measurements of DLBTF. Their narrower importance is methodological: canonical turbulence can be perturbed in controlled ways and the future response can be compared to an unperturbed or appropriately matched reference.

¹² M. P. Encinar and J. Jiménez, “Identifying causally significant features in three-dimensional isotropic turbulence,” *Journal of Fluid Mechanics* 965 (2023): A20, doi:10.1017/jfm.2023.407.

¹³ K. Osawa and J. Jiménez, “Causal features in turbulent channel flow,” *Journal of Fluid Mechanics* 1000 (2024): A4, doi:10.1017/jfm.2024.838.

The positive assay uses that possibility at the nonlinear-transfer interface itself. Small, normalized interventions are applied to declared transfer directions at a frozen reference state. The intervention is then released, and the subsequent β -relevant response is measured. The question is not merely whether the flow notices the intervention. It is how many independent directions of transfer alteration produce independently distinguishable, regime-relevant future consequences above the frozen uncertainty floor. That is the job assigned to Causal Transfer Response Rank.

No one representation is allowed to carry the result by fiat. Nonlinear transfer can be resolved through spectral or interaction-based decompositions and through physical-space or localized constructions.¹⁴ The protocol therefore requires at least two competent representations that expose the transfer burden differently. Their raw coordinates and raw ranks need not be identical. For identity-level use, the representation-level evidence must support the same transfer-support status, either directly or through a preregistered cross-representation mapping that preserves the physical comparison. If the representations cannot be reconciled under that rule, the status remains UNRESOLVED rather than being averaged into a verdict.

Growth alone is also insufficient. Repeating many copies of one bounded local transfer module can manufacture a larger raw rank without demonstrating a new kind of distributed load-bearing support. The enlargement test therefore asks whether the larger system contains β -relevant causal directions not already inherited from, rescaled from or replicated from the smaller system. The smaller response subspace is mapped into the larger one under the preregistered correspondence; only response content outside that inherited subspace can count as causal novelty. Normalization, nonfactorizability, cross-unit growth and an independent intensive check provide alternative routes for attacking the replication explanation.

Finally, the rank-bearing directions have to carry load. Suppressing the transfer subspace identified by the assay must produce a β -relevant deficit relative to a matched sham intervention. Restoration of the suppressed capacity, or a competence-matched replacement under the frozen rule, must recover the missing

¹⁴ A. Couteau, P. Dimopoulos Eggenchwil and P. Jenny, “The influence of energy-containing scales on the distribution of spectral energy transfers,” *Journal of Fluid Mechanics* 1032 (2026): A15, doi:10.1017/jfm.2026.11357; B. Yeung, T. Chu and O. T. Schmidt, “Triadic orthogonal decomposition reveals nonlinearity in fluid flows,” *Journal of Fluid Mechanics* 1031 (2026): A34, doi:10.1017/jfm.2026.11183; N. A. K. Doan, N. Swaminathan, P. A. Davidson and M. Tanahashi, “Scale locality of the energy cascade using real space quantities,” *Physical Review Fluids* 3 (2018): 084601, doi:10.1103/PhysRevFluids.3.084601.

function to the required degree. A direction that is mathematically distinguishable but whose removal has no relevant consequence is not identity-bearing merely because it contributes to a matrix rank.

These controls turn OPENING into a positive but finite experimental claim. The evidence must show that the lower bound on causally required transfer capacity grows beyond uncertainty across the declared physical enlargement ladder, that the added directions are not merely inherited or replicated, that the result survives competent alternative representations and robustness tests, and that the identified directions actually carry the reference-regime burden. The preregistered bounded-interface challenger search must also satisfy its coverage criterion, and no competent bounded construction may survive under the same β burden. That challenger search does not establish OPENING; the positive lower-bound evidence does. OPENING then means that the measured lower bound continues to increase across the powered ladder while the novelty, load-bearing and adequacy controls remain satisfied. It does not establish divergence under indefinite enlargement, rule out every finite model, or measure a global minimal model dimension.

The next step is therefore mathematical rather than rhetorical: define the causal response operator and show exactly what its rank can and cannot lower-bound.

7. Causal Transfer Response Rank

The transfer-side experiment requires a lower bound on physical causal capacity rather than a count of resolved degrees of freedom. For one independently warranted flow X , one physical enlargement level λ , one reference state x and one competent transfer representation R , let $a \in \mathbb{R}^p$ denote a normalized intervention applied to independently defined directions of the nonlinear-transfer term. Let $y \in \mathbb{R}^q$ denote the preregistered, normalized future response entering the β fidelity burden. The local intervention experiment defines a map

$$F^R(X, \lambda, x; a) = y,$$

with the unperturbed flow at $a = 0$. Its first-order causal response operator is

$$J^R(X, \lambda, x) = D_a F^R(X, \lambda, x; 0).$$

The dimensions p and q are properties of the chosen intervention and response descriptions. They are not themselves estimates of turbulent complexity. The object of interest is which independent changes to nonlinear transfer remain distinguishable in β -relevant future behavior.

The first lower bound follows directly from factorization. Suppose a proposed β -sufficient physical realization carries the effect of the intervention to the future response through m independent transfer-interface coordinates $z \in \mathbb{R}^m$. Locally,

$$a \rightarrow z(a) \rightarrow y(z).$$

If these maps are differentiable at the reference state, the chain rule gives

$$D_a y = D_z y \cdot D_a z.$$

Therefore

$$\text{rank}(D_a y) \leq m,$$

and hence

$$m \geq \text{rank}(J^R(X, \lambda, x)).$$

The result is modest but useful. Under the stated differentiable factorization, any β -sufficient physical realization whose intervention effect passes through m independent transfer-interface coordinates must satisfy $m \geq \text{rank}(J^R(X, \lambda, x))$. A large downstream state cannot manufacture additional first-order input directions after a genuinely lower-rank causal interface. Conversely, a large grid, a large latent representation or a large number of resolved velocity modes does not by itself increase this lower bound. The rank belongs to the intervention-to-response relation.

Exact rank is too brittle for an empirical criterion. The response operator is estimated with finite data, finite numerical accuracy, a finite intervention amplitude and a β tolerance that already permits some departure from the reference response. Let $\epsilon\beta$ denote the frozen total operator-level error allowance after the permitted causal-response discrepancy and the relevant estimation, numerical and intervention-linearization uncertainties have been accounted for. Let

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r \geq 0$$

be the singular values of the normalized response operator.

For the spectral norm, the optimal error of any rank- m approximation is σ_{+1} .¹⁵ It follows that if

$$\sigma_{+1} > \epsilon\beta,$$

then no rank- m linear response operator can approximate the measured normalized response operator within the frozen tolerance. The β -tolerant Causal Transfer Response Rank is therefore the number of singular directions that remain resolvable above the frozen error floor,

$$\text{CTRR}_{\epsilon\beta}(X, \lambda, x, R) = \#\{i : \sigma_i > \epsilon\beta\}.$$

Because the tolerance-aware count depends on normalization, the scaling of intervention and response coordinates is part of preregistration. Equivalent-basis normalization checks are required so an apparent change in CTRR cannot be manufactured by rescaling coordinates while leaving the underlying response geometry unchanged.

Combining this approximation bound with the differentiable-factorization assumption, any β -sufficient physical transfer realization within the declared regularity class must carry at least $\text{CTRR}_{\epsilon\beta}$ independent causal transfer-interface dimensions at that state. This is the positive lower-bound step missing from an unsuccessful closure search. The inference no longer depends on how many candidate models investigators happened to try. It is carried by the response measured from the flow.

Several restrictions are important. CTRR is local to the declared intervention family and observation horizon. It does not measure the dimension of the complete Navier–Stokes state, an attractor, an inertial manifold or a globally minimal turbulence model. Its lower-bound interpretation also presumes a physically competent interface: a proposed closure cannot evade the count by encoding arbitrarily complex information discontinuously into one scalar or by relying on infinitely precise parameter tuning. The admissible comparison class must remain continuous and sufficiently regular over the calibrated intervention range for its causal response to be scientifically stable. That regularity class is a stated scope condition of the CTRR

¹⁵ C. Eckart and G. Young, “The Approximation of One Matrix by Another of Lower Rank,” *Psychometrika* 1, no. 3 (1936): 211–218, doi:10.1007/BF02288367.

inference. A physically credible challenger outside it is not silently excluded; it would require a separately specified assay.

Hidden states do not provide an automatic escape. If memory variables, stochastic internal states, learned coordinates, delayed variables or auxiliary equations create independent routes by which the one-shot transfer intervention affects the future β response, those routes belong on the transfer side of the causal cut. They need not be counted simply because they exist; they have to be counted insofar as they carry independent causal response. If a strongly memory-dependent candidate cannot be tested adequately by the one-shot operator, that limitation must be identified during method development or calibration. A multi-time response construction may then be preregistered before decisive data are inspected. If the inadequacy becomes apparent only during decisive adjudication, the affected burden remains unresolved or the execution is invalid rather than being repaired post hoc.

The state-level rank is not yet the regime-level result. At each enlargement level, the experiment samples a preregistered ensemble of β -qualified reference states and estimates CTRR with uncertainty for each. The regime-level lower bound is then fixed as the largest integer ℓ for which the required fraction of β -qualified states has a lower confidence bound of at least ℓ . The required fraction and confidence rule are set during calibration. One exceptionally high-rank state cannot establish OPENING, and an average cannot be chosen so that systematically high-rank states disappear.

The decisive quantity is how this lower bound behaves under admissible physical enlargement. If $L\beta^R(X, \lambda)$ denotes the resulting regime-level lower bound in representation R, OPENING requires supported growth of $L\beta^R$ across the preregistered enlargement ladder beyond its uncertainty envelope. A statistically powered plateau blocks an OPENING verdict but does not by itself establish BOUNDED; BOUNDED still requires the competent bounded-interface result defined above. Growth that appears only after numerical refinement, without physical enlargement, does not count. Neither does growth produced solely by duplicating an already bounded causal module.

The result must also survive the second competent representation and the novelty and load-bearing controls described above. The larger system must exhibit β -relevant response directions outside the preregistered image of the smaller-system response

subspace, and suppressing the rank-bearing directions must impair the reference-regime burden relative to the matched sham. Only after those conditions are satisfied can CTRR growth contribute to an OPENING verdict.

CTRR therefore supplies a lower bound, not an asymptotic theorem. An observed sequence

$$L^{\beta^R}(X, \lambda_1) < L^{\beta^R}(X, \lambda_2) < \dots$$

over a competent finite ladder can support continuing opening over that ladder. It cannot prove

$$\lim_{\lambda \rightarrow \infty} L^{\beta^R}(X, \lambda) = \infty.$$

That stronger statement is unnecessary for the proposed identity test. For this protocol, the transfer-support question is explicitly three-way: competent evidence may support BOUNDED, support OPENING over the declared powered enlargement ladder, or leave the relation UNRESOLVED. The experiment is designed to answer that finite-scope question and no stronger one.

8. Testing the history side

The transfer test does not decide whether the regime is dynamically exhausted by one thin recurrent history family. That is the separate burden assigned to History-Family Nondegeneracy.

The HFN experiment begins from the same principle used on the transfer side: the object is fixed before the outcome is known. For each independently warranted flow, the physical regime, reference-state ensemble, β fidelity rule, perturbation neighborhood, observation horizon, symmetry, phase and time-shift quotient, recurrence-template class and history decision rule are fixed before decisive histories are inspected. The question is then whether nearby β -qualified realizations produce genuinely distinct histories while remaining members of that regime.

Distinct initial states are not enough. Two trajectories related only by a permitted spatial symmetry, phase displacement or time translation should not count as separate histories merely because their coordinates differ. Those equivalences are removed under the frozen quotient before nondegeneracy is assessed. Nor is ordinary sensitive

dependence sufficient by itself. An unstable periodic orbit may possess expanding directions and still remain one periodic orbit. HFN asks whether the realized regime supports histories that separate into genuinely distinct admissible continuations rather than merely moving along, returning to or trivially transforming one dynamically thin recurrent template.

The positive burden therefore combines several ideas that are weaker when taken separately. There must first be a realized family of histories rather than one selected trajectory. Multiple members of that family must remain within the same β -qualified regime while separating dynamically. The separation must arise through the regime's own dynamics rather than being reducible to an externally imposed label or to the random seed of a forcing process. The resulting histories must not be exhausted, after the frozen quotient, by the recurrent template class against which the assay was calibrated. And the distinction must persist long enough to remain a regime-level fact rather than a transient perturbation that immediately collapses back onto one thin itinerary.

Operationally, each flow case begins with a preregistered neighborhood of β -qualified reference states and at least two competent history representations. Histories are propagated over the frozen horizon, the declared symmetry, phase and time-shift quotient is applied, and the recurrence rule asks whether same-regime continuations can all be absorbed into the frozen recurrent-template class. NONDEGENERATE requires a same-regime branching witness that survives both history representations and the frozen recurrence rule; unresolved representation disagreement remains UNRESOLVED.

This does not require an exhaustive catalogue of every unstable periodic orbit or recurrent solution. Such a requirement would make the positive burden depend on proving that no undiscovered orbit exists. Instead, a competent same-regime branching witness can establish nondegeneracy if the preregistered recurrence diagnostics show that the observed histories cannot all be absorbed into the frozen thin recurrent class. Exhaustive enumeration is unnecessary; adequate discrimination is required.

The negative verdict is correspondingly strong. RECURRENT-THIN is not assigned merely because branching was difficult to observe. It requires positive evidence, under

an adequately powered assay, that the β -qualified history family is exhausted by the declared dynamically thin recurrent or quasirecurrent class at the predeclared level and horizon. It is therefore a scope-bounded empirical verdict, not a proof about every conceivable history beyond the frozen assay. If the method cannot distinguish those alternatives, the status is UNRESOLVED.

Lyapunov analysis can contribute to this experiment without defining it. Full Lyapunov spectra have been computed directly for forced three-dimensional homogeneous isotropic turbulence,¹⁶ and leading Lyapunov-vector dynamics have been studied in turbulent channel flow.¹⁷ Such methods demonstrate that perturbation growth and history sensitivity can be interrogated in canonical three-dimensional turbulence. They do not establish HFN by themselves. A positive Lyapunov exponent says that nearby trajectories separate locally; it does not say whether the resulting regime histories escape exhaustion by the frozen recurrent class.

The same firewall applies to stochastic forcing. Forced turbulence may be maintained with stochastic inputs, but many realized histories generated by many forcing realizations do not automatically demonstrate endogenous history-family nondegeneracy. The experiment must distinguish variation attributable solely to differing forcing histories from branching that survives the preregistered endogenous-variation criterion. Randomness can be part of the physical realization without being allowed to answer the identity question by itself.

The recurrent-solution literature makes the boundary case particularly important. An isolated unstable periodic solution may carry nonlinear transfer, exhibit unstable directions and reproduce striking turbulent statistics while remaining recurrent by construction. The containing turbulent regime can have a different HFN status because its trajectories may approach, shadow, leave and revisit multiple recurrent structures while remaining within the same regime. The experiment therefore fixes the level before inspection. An isolated orbit may return RECURRENT-THIN while the containing regime returns NONDEGENERATE without contradiction. What is forbidden is switching between those levels after seeing which answer protects the criterion.

¹⁶ M. Hassanali and V. Raman, “Lyapunov spectrum of forced homogeneous isotropic turbulent flows,” *Physical Review Fluids* 4 (2019): 114608, doi:10.1103/PhysRevFluids.4.114608.

¹⁷ N. Nikitin, “Characteristics of the leading Lyapunov vector in a turbulent channel flow,” *Journal of Fluid Mechanics* 849 (2018): 942–967, doi:10.1017/jfm.2018.418.

For each preregistered flow case, HFN therefore returns one of three statuses. NONDEGENERATE means that a competent, powered history assay establishes a realized same-regime family whose members separate through the registered endogenous dynamics, remain distinct after the frozen quotient, are not exhausted by the declared thin recurrent class, and persist as distinct continuations over the predeclared regime-level horizon. RECURRENT-THIN means that competent evidence establishes recurrence exhaustion at the predeclared level and horizon under the same representation, power and recurrence-adequacy requirements. UNRESOLVED means that neither positive claim has been earned.

This three-way structure matters because HFN is a necessary conjunct. Failure to establish NONDEGENERATE cannot be counted as evidence against an independently warranted turbulent flow; it may simply expose an inadequate recurrence horizon, weak perturbation design, insufficient statistical power or an unresolved history representation. Equally, failure to demonstrate recurrence collapse cannot be converted into a positive HFN result. The two scientific claims have separate evidentiary burdens.

The transfer and history assays now meet the same standard of falsifiability while remaining distinct. DLBTF asks whether sufficient causal transfer support closes or continues to open under physical enlargement. HFN asks whether realized regime histories remain nondegenerate or collapse onto a dynamically thin recurrent family. Either can fail while the other survives. The identity candidate earns no protection from that possibility; it depends on it.

9. Freezing the question before decisive data

Method-development data may be used to discover unstable interventions, uninformative observables, inadequate recurrence diagnostics or incompetent representations; they do not contribute decisive evidence.

Calibration then freezes the β mappings and tolerances, intervention and normalization rules, CTRR error floor and enlargement ladder, representation-competence tests, HFN neighborhood, horizon, quotient and recurrent-template class, ensemble and power rules, restoration tests and robustness requirements. Once decisive adjudication begins, that repair window closes.

The protected internal-wall and free-shear holdouts are then opened under those frozen rules to test portability beyond the primary channel and round-jet implementations. If a holdout reveals that the frozen method is unusable, that execution is invalid and a new independent holdout is required; the exposed data may not be used to retune the same adjudication.

The same information order fixes scope, physical level and comparator standing. Positives are not removed after falsification, negatives are not reclassified after passing, and recurrent objects do not switch levels to rescue the candidate. This is part of the scientific content of the claim: the intended result is allowed to lose.

10. Compound adjudication at one flow

DLBTF and HFN are evaluated independently and combined only after both assays have returned their scientific statuses.

For an independently warranted turbulent regime, a compound pass requires the complete DLBTF assay to pass with transfer-support status OPENING and the complete HFN assay to pass with history-family status NONDEGENERATE. On the transfer side, realized-history standing, nonlinear-transfer involvement, load-bearingness and causal continuation must all be established, and the result must survive representation neutrality, bounded-interface search adequacy, power, enlargement robustness, anti-replication, causal-novelty, suppression and restoration controls. On the history side, the five HFN jobs must all be established and the result must survive representation, power and recurrence-exhaustion adequacy. A status label without those controls does not carry identity weight.

A positive flow returning BOUNDED defeats DLBTF necessity whether or not its history family is nondegenerate. A positive flow returning RECURRENT-THIN defeats HFN necessity whether or not its transfer support opens. Because the candidate is conjunctive, one demonstrated necessary-conjunct failure is enough.

UNRESOLVED behaves differently. If an independently warranted turbulent flow returns OPENING on the transfer side but UNRESOLVED on the history side, the identity claim has not failed there, but it has not passed either. The same is true of NONDEGENERATE combined with unresolved transfer support. Lack of a

positive result cannot be converted into a negative result merely to finish the classification.

The logic reverses appropriately for independently warranted nonturbulent comparators. A negative case is successfully excluded if competent evidence establishes at least one necessary-conjunct failure: BOUNDED transfer support, RECURRENT-THIN history, or, for a synthetic false substitute, failure of the required physical provenance. It does not have to fail both DLBTF and HFN. Requiring both would incorrectly treat the conjunction as though every negative had to differ from turbulence in the same way.

This matters for the deliberately heterogeneous negative roster. A periodic or steady coherent solution may carry rich nonlinear transfer while being excluded by recurrence collapse at the isolated-object level. A low-dimensional chaotic Taylor–Couette case may possess a nondegenerate history family while still being excluded if its causally sufficient transfer architecture closes to a bounded support. A synthetic statistical mimic may reproduce distributions or apparent history diversity while failing the physical provenance needed for the declared Navier–Stokes identity claim. Different false substitutes are allowed to fail for different reasons.

The dangerous result is the opposite one. If an independently warranted nonturbulent comparator competently returns both OPENING and NONDEGENERATE while satisfying the relevant physical and evidentiary conditions, the criterion has failed sufficiency. That is not a marginal anomaly to be explained away. It shows that the proposed conjunction admits something outside the class it was supposed to discriminate.

The recurrent boundary pair is adjudicated separately because it tests whether the criterion respects physical level. The isolated recurrent solution and the containing turbulent regime retain their frozen identities throughout the experiment. The isolated solution may legitimately be RECURRENT-THIN while the containing regime is NONDEGENERATE. That pattern supports the intended level discrimination. If a clean verdict can be obtained only by switching the same case between orbit and regime descriptions after the results are known, the execution is invalid.

The two holdouts carry the same necessity burden as the core positive flows once decisive adjudication begins. They are not optional confirmations. A candidate that

passes its development families but returns BOUNDED or RECURRENT-THIN in a required within-family holdout has failed the claimed Σ_C necessity. A holdout that remains unresolved prevents class-level closure until the relevant burden is competently adjudicated.

Only after these flow-level records are fixed is the global identity claim assembled.

11. Closing, breaking, or leaving the identity claim open

The global adjudication has three outcomes:

CLOSED WITHIN Σ_C

NOT CLOSED WITHIN Σ_C

NOT YET ADJUDICABLE WITHIN Σ_C

These outcomes deliberately separate contradiction from insufficient evidence.

The identity claim is NOT CLOSED when a competent result defeats the proposed identity criterion at a declared closure burden. One independently warranted positive or required holdout returning BOUNDED defeats DLBTF necessity. One returning RECURRENT-THIN defeats HFN necessity. One independently warranted nonturbulent comparator competently satisfying the full compound criterion defeats sufficiency. An earned redundancy challenge also returns NOT CLOSED when a simpler established criterion absorbs the complete discrimination job assigned either to the claimed transfer-side identity burden or to the full DLBTF $\wedge$ HFN conjunction. An earned residual-identity challenge returns NOT CLOSED when an independent same-level identity burden survives the declared absorption tests and changes the relevant classification. These redundancy and residual routes are identity-level adverse results; they are not retroactively relabeled as per-flow BOUNDED or RECURRENT-THIN results. A boundary result that survives only through post-result level switching invalidates that route rather than supporting closure.

NOT YET ADJUDICABLE applies when no NOT CLOSED result has been earned but the scientific question remains open because at least one required burden is unresolved, inadmissible or execution-invalid. An underpowered CTRR spectrum,

unresolved disagreement between competent transfer representations, inadequate history discrimination, an unusable decisive holdout, failure to establish recurrence exhaustion, unresolved external-mimic provenance, an unresolved redundancy or residual audit, or another unresolved required burden cannot be counted as either success or falsification. A frozen-scope, frozen-role, frozen-level or pre-outcome-integrity failure invalidates the affected execution route and likewise cannot support closure. The absence of closure is not itself evidence against the criterion.

CLOSED carries the strongest burden. Every independently warranted positive core family must satisfy both conjuncts under the frozen assay. The required internal-wall and free-shear holdouts must do the same. Every negative or false-substitute case must be competently excluded by at least one necessary burden or the appropriate provenance rule. The isolated recurrent object and containing regime must retain coherent level-specific adjudications. The transfer result must survive representation, bounded-interface search coverage, power, enlargement robustness, causal-novelty, anti-replication, suppression and restoration pressure. The history result must survive its corresponding representation, power, recurrence and boundary tests.

Even that partition is not sufficient by itself. Before an identity claim closes, the proposed conjunction must face two final attacks.

The first asks whether a simpler established criterion already performs the complete discrimination job. If another criterion includes the same positive and holdout families, excludes the same difficult negatives, respects the same recurrent boundary, and survives the same causal and enlargement pressures without needing both DLBTF and HFN, then the compound proposal has not earned an independent identity role merely by redescribing the partition.

The second asks whether an independent identity burden is still missing. A candidate third property matters only if it performs a distinct discrimination job that DLBTF and HFN do not already perform, has its own physical failure condition at the same level, changes the classification of a relevant case, and cannot be absorbed into the existing criterion. A surviving burden of that kind prevents closure. An additional descriptive feature that tracks the same distinction does not.

Only after those rival and residual challenges have been discharged can the strongest result be stated:

Within Σ_C , the preregistered evidence supports Distributed Load-Bearing Transfer Flow together with History-Family Nondegeneracy as the necessary-and-sufficient operational identity criterion for the declared canonical class of three-dimensional incompressible Newtonian turbulence.

In compressed form:

Turbulence identity closed within Σ_C .

That sentence is conditional on an empirical result that has not yet been produced. The present work specifies the experiment by which it can become true, become false, or remain scientifically open.

The falsifiability is therefore asymmetric in a useful way. A single competent counterexample can break necessity or sufficiency. Closure requires the entire declared burden to survive. Uncertainty does neither.

That asymmetry is intentional. An identity criterion should be difficult to establish and easy to kill when the physics disagrees.

12. Can the experiment be executed?

The proposed criterion would be of little scientific use if its falsifiers were clear in principle but inaccessible in practice. Existing turbulence work does not establish DLBTF, HFN or their conjunction, but the preceding sections already document the constituent capabilities the test requires: high-fidelity realizations of the positive flow families, controlled causal perturbations, multiple nonlinear-transfer representations, and Lyapunov or history-sensitive diagnostics. Those precedents support feasibility, not validation.

The holdouts are similarly grounded in real physical variation. Pipe, channel and boundary-layer turbulence exhibit important common structure alongside measurable flow-specific differences, which is precisely why one internal-wall realization should

not silently stand for the entire canonical wall-bounded family.¹⁸ Jet and wake studies likewise exhibit distinct turbulent/non-turbulent interface structures within free-shear turbulence.¹⁹ The holdouts exploit those differences as portability tests rather than pretending that each geometry defines a different turbulence identity.

What remains difficult is the integration. Specialists still have to choose and justify the actual physical enlargement parameter for each flow, instantiate the four β fidelity jobs in flow-appropriate observables, select competent transfer representations, calibrate intervention amplitudes and error floors, construct statistically powered state ensembles, define the recurrence quotient and recurrent-template class, and establish which perturbation horizon genuinely distinguishes history-family branching from transient divergence. Those choices cannot be supplied by the identity criterion itself. They are turbulence-domain work.

That dependence on specialist judgment is not a loophole if the judgment is exercised before decisive outcomes are known. The protocol does not require every flow to share identical observables, identical coordinates or identical numerical machinery. It requires the same scientific jobs to be performed under frozen, auditable rules. A channel-flow wall-stress response and a round-jet downstream development measure need not look alike to carry the same fidelity burden.

Nor does feasibility require every component to succeed on the first implementation. An intervention family may prove poorly conditioned. A proposed representation may fail its competence test. A recurrence horizon may be too short to discriminate the relevant histories. A physical enlargement ladder may be underpowered. Those are legitimate outcomes of method development. What they cannot become is evidence for OPENING, BOUNDED, NONDEGENERATE or RECURRENT-THIN merely because a desired measurement was hard to obtain.

The practical claim is therefore intentionally limited. The constituent experimental ideas—canonical high-fidelity flows, causal perturbation, nonlinear-transfer decomposition, recurrent-solution analysis, history sensitivity and cross-family

¹⁸ J. P. Monty, N. Hutchins, H. C. H. Ng, I. Marusic and M. S. Chong, “A comparison of turbulent pipe, channel and boundary layer flows,” *Journal of Fluid Mechanics* 632 (2009): 431–442, doi:10.1017/S0022112009007423.

¹⁹ J. Westerweel, J. M. Pedersen, C. Fukushima and J. C. R. Hunt, “Mechanics of the turbulent/non-turbulent interface of a jet,” *Physical Review Letters* 95 (2005): 174501, doi:10.1103/PhysRevLett.95.174501; D. K. Bisset, J. C. R. Hunt and M. M. Rogers, “The turbulent/non-turbulent interface bounding a far wake,” *Journal of Fluid Mechanics* 451 (2002): 383–410, doi:10.1017/S0022112001006759.

comparison—already exist in turbulence research. The proposed contribution is to place them under one preregistered discrimination problem with common failure logic. Whether that integration works well enough to adjudicate the identity claim remains an empirical question.

13. What a closed result would establish

If every required burden closes, the result would settle the operational identity question posed in this paper within Σ_C . It would support DLBTF together with HFN as a necessary-and-sufficient criterion for the declared canonical class: the independently warranted turbulent regimes would satisfy both burdens, the independently warranted false substitutes would be excluded by at least one of them or by the required physical provenance, the recurrent boundary would remain coherent, and the criterion would survive its representation, robustness, holdout, redundancy and residual challenges.

That would be a claim about what distinguishes the declared turbulent regime family, not a claim that every problem involving turbulent flow had disappeared. CLOSED is an empirical protocol disposition, not a mathematical proof over every untested Navier–Stokes realization in Σ_C . It means that the preregistered evidence supports the class-level operational criterion under the declared roster, holdouts and robustness tests; later competent counterevidence would reopen that conclusion.

It would not solve the mathematical regularity problem for the three-dimensional Navier–Stokes equations. It would not provide closed-form solutions of arbitrary turbulent flows. It would not supply every engineering closure, predict every turbulent statistic, determine every transition threshold, explain every intermittency exponent or replace the large literature on turbulent mechanisms and structures. Those are different questions.

Nor would closure within Σ_C automatically settle the identity of compressible, rotating, stratified, buoyant, multiphase, reactive, magnetohydrodynamic, plasma, quantum or superfluid turbulence. Those regimes were excluded from the first claim precisely so that their additional physics could not be hidden inside an overbroad word. A later extension would have to show that the criterion translates competently to the new domain or identify the additional burden that prevents it from doing so.

A DLBTF result would also remain narrower than a theorem of infinite complexity. OPENING over a finite, powered enlargement ladder does not prove that CTRR diverges as some parameter tends to infinity. It does not prove that no finite turbulence model can ever exist. A finite model could remain useful, predictive or even remarkably compact while failing to supply a bounded physical transfer interface that preserves the complete β causal burden across the tested enlargement ladder.

Similarly, HFN closure would not say that turbulence is “random” in an unrestricted sense or that exact coherent structures cease to matter. It would support the narrower conclusion that, under the preregistered history assay, the independently warranted regime is not exhausted by the declared thin recurrent itinerary or finite thin family; recurrent structures may remain dynamically central.

14. The specialist task

The present proposal does not ask turbulence specialists to accept DLBTF and HFN as established properties of turbulence. It gives them explicit ways to make the proposal fail.

A specialist who suspects that DLBTF is unnecessary does not have to defeat every conceivable transfer description. One competent bounded physical interface that preserves the preregistered β burden across a required positive enlargement ladder is enough to break transfer necessity.

A specialist who suspects that HFN is unnecessary can attack the history side directly. Competent evidence that one required turbulent regime is RECURRENT-THIN at the preregistered regime-family level breaks history necessity.

A specialist who thinks the conjunction is too permissive can attack sufficiency instead. Produce an independently warranted nonturbulent comparator that survives the same fidelity discipline and competently returns both OPENING and NONDEGENERATE. The criterion then admits something it was supposed to exclude.

A specialist who thinks the two-conjunct construction is unnecessarily complicated can supply a simpler criterion that performs the entire discrimination job: positives,

holdouts, difficult negatives, recurrent boundary, causal tests and physical enlargement included. If it genuinely absorbs the work, the compound criterion should give way.

And a specialist who thinks the proposal still misses something can identify a third independent identity burden. It has to do more than name another familiar property of turbulence. It must perform a discrimination job not already carried by DLBTF or HFN, possess its own same-level physical failure condition, change the classification of a relevant case, and survive an attempt to absorb it into the existing criterion.

That is the wager of the experiment. The criterion is not protected by the difficulty of the turbulence problem. It is exposed to it.

The burden is not to agree with the proposed identity. The burden is to make the physics answer it.

15. Conclusion

This paper isolates one version of the turbulence problem: whether a declared canonical class can be identified by a criterion that is necessary across accepted turbulent regimes, sufficient against difficult neighboring cases, and vulnerable to decisive counterexample.

The proposed candidate is $DLBTF \wedge HFN$. Neither conjunct is established here as a fact about turbulence. What is supplied is a preregisterable empirical test that can support the conjunction, defeat it, or leave a specific burden unresolved without moving the scope or decision rules after the result.

If the complete burden survives, the licensed conclusion is turbulence identity closed within Σ_C under the proposed operational criterion. If the physics supplies a competent counterexample, the criterion fails where it fails.

The aim is not to settle turbulence by definition. It is to make this old identity question answerable in a form specialists can attack.

The criterion does not need agreement.

It needs a verdict.

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The Turbulence Identity Problem

A Falsifiable Experimental Criterion

Signed,

Rick